

Quote of the day: “*‘Math is hard,’ Barbie famously declared. Well, Barbie was right, but math is not uniquely hard. Playing the violin is hard, hitting a baseball is hard, and learning a second language is hard. What seems to make mathematics different from playing the violin or learning Chinese is that the struggle to play violin doesn’t make people feel defeated.*” – Stephen Kennedy

Remarks: This is a review sheet for the midterm on Friday, which will take place during class time in our usual location. The midterm will include questions on set equality, injectivity/surjectivity, and induction. There will be four questions, with one of the topics (likely injectivity/surjectivity) appearing twice. On this practice sheet, you’ll find worked out examples of each of the three types of problems, along with a practice problem for each type. We’ll go over the worked out examples during class.

Exercise 1. Prove that $\{x \in \mathbb{R} : x^2 \leq 1\} = [-1, 1]$ by double containment.

Proof. ($\subseteq$): Let $a \in \{x \in \mathbb{R} : x^2 \leq 1\}$. Then $a \in \mathbb{R}$ and $a^2 \leq 1$, so that $(1 - a)(1 + a) = 1 - a^2 \geq 0$. Then $a \in \mathbb{R}$, so it suffices to show that $|a| \leq 1$. We have that $\sqrt{a^2} = |a|$, so taking the squareroot of both sides of the inequality $a^2 \leq 1$ yields $|a| = \sqrt{a^2} \leq \sqrt{1} = 1$. Hence $a \in [-1, 1]$.

($\supseteq$): Let $a \in [-1, 1]$. Then $|a| \leq 1$, so we have $a^2 = |a|^2 \leq 1$, so $a \in \{x \in \mathbb{R} : x^2 \leq 1\}$. $\square$

Exercise 2. Let $A \subseteq X$ and $B \subseteq X$ be sets. Prove that $X \setminus (A \cup B) = (X \setminus A) \cap (X \setminus B)$

Proof. $\square$

Exercise 3. Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^3$. Is f injective? What is its image? Is it surjective? Prove your claims.

Proof. The function f is injective. Let $a, b \in \mathbb{R}$, and suppose $a^3 = b^3$. Taking the cubed root of both sides, we obtain $a = b$.

The function f is surjective, so its image is $\mathbb{R}$. Let $y \in \mathbb{R}$. Then $\sqrt[3]{y} \in \mathbb{R}$, and $\sqrt[3]{y}^3 = y$, so f is surjective. We proved in class that a function is surjective if and only if its image is all of the codomain, so f surjective implies $f(\mathbb{R}) = \mathbb{R}$. $\square$

Exercise 4. Consider the absolute value function $|\cdot|: \mathbb{R} \rightarrow \mathbb{R}$. Is $|\cdot|$ injective? What is its image? Is it surjective? Prove your claims.

Proof. $\square$

Exercise 5. Prove that for all $n \geq 4$, we have $3n < 2^n$.

Proof. **Base case:** The base case is when $n = 4$. In this case, we have $3 \cdot 4 = 12 < 2^4 = 16$, so the statement is true.

Induction step: Suppose the statement is true for some $n \geq 4$. That is, assume $3n < 2^n$. We will show that $3(n+1) < 2^{n+1}$. We have $3(n+1) = 3n + 3$, and by assumption, $3n + 3 < 2^n + 3$. Since $n \geq 4$, $2^n + 3 < 2^n \cdot 2$, so

$$3(n+1) = 3n + 3 \leq 2^n + 3 < 2^n \cdot 2 = 2^{n+1}.$$

Therefore, by induction, the statement is true for all $n \geq 4$. □

Exercise 6. Prove by induction that

$$\sum_{k=0}^n 2^k = 2^{n+1} - 1$$

for all $n \in \mathbb{N}$.

Proof. □